VISUALIZATION OF BED OCCUPANCY USING TIME-SERIES DASHBOARDS AND PEAK LOAD COMPARISON CHARTS

A REAL-TIME RESEARCH PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING(AI&ML)

Under the Guidance of

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ACCREDITED TO NAAC & NBA(CSE, IT,EEE, ECE)

Kistapur (V),Medchal(Dt)-501401

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI&ML)

Malla Reddy Engineering College and Management Sciences

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This is to certify that the project report titled,"**VISUALIZATION OF BED OCCUPANCY USING TIME-SERIES DASHBOARDS AND PEAK LOAD COMPARISON CHARTS**" is a bonafide record of the project work done by

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The award of Bachelor of Technology in Department Of Computer Science And Engineering (Ai&Ml) the academic year 2025-2026.

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We, undersigned, hereby declare that the project report **"VISUALIZATION OF BED OCCUPANCY USING TIME-SERIES DASHBOARDS AND PEAK LOAD COMPARISON CHARTS",** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the Jawaharlal Technological University, Hyderabad is a Bonafide work done by me under the guidance of **Mrs. HEENAN KAUSAR M.tech** Department of **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI&ML).**
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PEO3	MREM B.Tech CSM graduates shall be able to handle societal and environmental problems with ethical values as demanded by the society.

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CO2	Undertake problem identification, formulation and solution.
CO3	Design engineering solutions to complex problems utilizing a systems approach.
CO4	Work with practicing engineers
CO5	Demonstrate the knowledge and skills acquired during the course work

ABSTRACT

This project studies hospital capacity data to understand how hospital beds and ICU beds are used in different regions and time periods. The dataset contains information such as total hospital beds, available beds, hospital beds needed, ICU beds, and occupancy rates. These details help us understand how hospitals handle patient demand and whether they have enough resources to treat patients. In this project, Exploratory Data Analysis (EDA) was used to study the dataset. Different charts and visualizations such as line charts, bar charts, histograms, scatter plots, pie charts, box plots, and 3D graphs were created to better understand the data. An interactive dashboard was also developed to show important information like hospital bed demand, ICU demand, and occupancy rates.From the analysis, we can see that hospital bed usage is high in some regions. In certain cases, the number of beds needed is close to or higher than the number of available beds. This means hospitals may face problems if the number of patients increases. ICU beds are also very important because they are used for serious patients. Overall, this project shows how data analysis and visualizations can help understand hospital capacity and healthcare demand. The results can help hospitals and healthcare planners manage resources better and prepare for future healthcare needs.

Keywords: Hospital Beds, ICU Beds, Occupancy Rate, Data Analysis, Data Visualization, Healthcare Planning, Exploratory Data Analysis (EDA), Time-Series Dashboards, Peak Load Comparison.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
EDA	Exploratory Data Analysis
ICU	Intensive Care Unit
HRR	Hospital Referral Region
CSV	Comma-Separated Values
KPI	Key Performance Indicator
DFD	Data Flow Diagram
UML	Unified Modeling Language
ER	Entity-Relationship
ML	Machine Learning
AI	Artificial Intelligence
API	Application Programming Interface
GUI	Graphical User Interface
RAM	Random Access Memory
OS	Operating System
IDE	Integrated Development Environment

CHAPTER 1

INTRODUCTION

1.1 Introduction to the Project

Healthcare systems play an important role in providing medical services and ensuring that hospitals have enough resources to treat patients. One of the major challenges in healthcare management is maintaining a proper balance between available hospital resources and patient demand. Hospital capacity data helps in understanding how hospital beds and ICU beds are used in different regions and time periods. By studying this data, healthcare authorities can plan resources better and improve patient care.

The hospital capacity dataset used in this project contains information about hospital resources across different regions. It includes variables such as total hospital beds, available hospital beds, hospital beds needed, total ICU beds, available ICU beds, ICU beds needed, and bed occupancy rates. These variables help in understanding how hospitals manage their capacity and how healthcare demand changes over time.

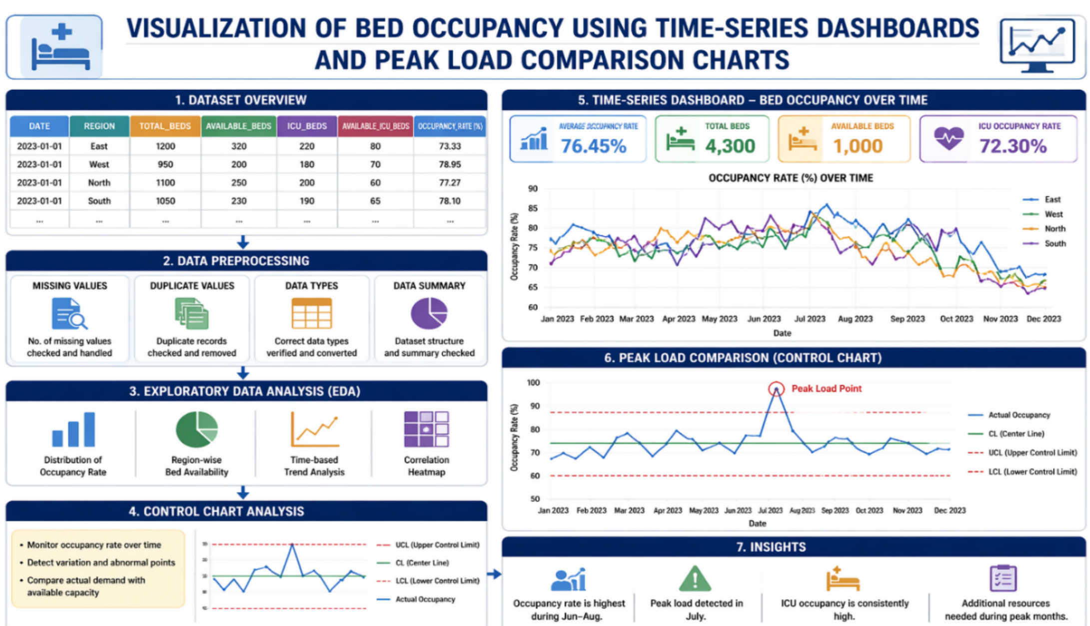

Fig. 1.1: Hospital Resources Overview

This project focuses on analyzing the dataset using Exploratory Data Analysis (EDA) and visualization techniques in Python. Different statistical methods and visualizations are used to study patterns and relationships between hospital beds, ICU demand, and occupancy rates. Several plots such as line charts, bar charts, scatter plots, histograms, box plots, pie charts, treemaps, and 3D visualizations were created to better understand the data.

1.2 Background of the Study

The management of hospital resources is a critical aspect of modern healthcare systems. With rapidly growing populations and increasing healthcare demands, hospitals are constantly challenged to allocate their limited resources efficiently. Hospital beds and ICU facilities represent two of the most valuable and limited resources within any healthcare institution.

Historically, hospital resource management relied on manual processes and paper-based recording systems. These approaches were not only time-consuming but also prone to human errors and lacked the ability to provide real-time insights. As healthcare data became more digitized, the need for sophisticated analytical tools grew significantly.

The advent of big data analytics and visualization technologies has opened new avenues for healthcare administrators to monitor, analyze, and predict resource utilization. Tools such as Python's Pandas and Plotly libraries, combined with comprehensive datasets, allow for in-depth exploration of healthcare patterns and trends.

This project builds upon this foundation by applying Exploratory Data Analysis (EDA) to a hospital bed occupancy dataset spanning multiple regions and time periods. The dataset includes key metrics such as bed occupancy rates, ICU utilization, and regional demand variations, providing a rich basis for analysis.

1.3 Problem Statement

Hospitals need enough beds, ICU units, and medical resources to treat patients properly. However, patient demand can increase due to population growth, diseases, or health emergencies. When hospitals do not have enough beds or ICU facilities, it can create problems such as overcrowding, delayed treatment, and pressure on healthcare staff. The dataset used in this project contains information about hospital capacity in different regions and time periods. It includes variables such as total hospital beds, available hospital beds, hospital beds needed, total ICU beds, available ICU beds, ICU beds needed, and bed occupancy rates. These variables help to show how hospitals manage their resources and how the demand for healthcare services changes over time.

The main problem addressed in this project is to analyze hospital bed utilization and identify possible shortages in hospital and ICU capacity. In some regions, the number of beds required may be close to or even higher than the number of beds available. This situation can affect the ability of hospitals to provide proper healthcare services.

1.4 Objective of the Project

The primary objective of this project is to analyse hospital capacity data and understand the patterns, demand, and utilization of hospital resources such as hospital beds and ICU beds. The specific objectives are as follows:

1. To perform Exploratory Data Analysis (EDA) on the hospital capacity dataset in order to understand its structure, variables, and statistical characteristics.
2. To analyse the statistical properties of hospital data, including measures such as mean, median, standard deviation, minimum, and maximum values to understand the distribution and variability of hospital bed demand and occupancy rates.
3. To study hospital bed utilization and examine how hospital bed demand changes across different time periods such as 6 months, 12 months, and 18 months.
4. To investigate the relationship between hospital beds and ICU beds in order to understand how healthcare resources are distributed and used across different regions.
5. To identify patterns and trends in hospital capacity using visualization techniques and statistical analysis.
6. To create different types of visualizations, including line charts, histograms, scatter plots, box plots, pie charts, treemaps, sunburst charts, and 3D plots to better understand hospital capacity and healthcare demand.
7. To interpret the results obtained from the analysis and draw meaningful conclusions about hospital bed demand, ICU capacity, and healthcare resource utilization.

8. To develop an interactive dashboard that visually presents hospital capacity information in an easy and understandable format.
9. To provide insights into healthcare resource planning that may help improve hospital management and decision-making.
10. To demonstrate the use of data analysis tools such as Python, Pandas, and Plotly for analysing healthcare datasets effectively.

1.5 Scope of the Project

The scope of this project is focused on the analysis and visualization of a hospital bed occupancy dataset containing records from multiple Hospital Referral Regions (HRRs) across three time periods: 6 months, 12 months, and 18 months. The project covers the following areas:

- Data preprocessing and cleaning using Python's Pandas library.
- Statistical analysis of hospital bed demand, ICU utilization, and occupancy rates.
- Visualization of healthcare data using Plotly Express, Plotly Graph Objects, and Matplotlib.
- Development of an interactive dashboard consolidating key performance indicators.
- Identification of regional differences and temporal patterns in hospital capacity.
- Analysis of the relationship between general hospital bed demand and ICU bed demand.

The project does not include predictive modeling, real-time data integration, or patient-level clinical data analysis. The focus remains on descriptive and exploratory analysis of the provided dataset.

1.6 Organization of the Project

This report is organized into six chapters.

- Chapter 1 provides the introduction, background, problem statement, objectives, and scope of the project.
- Chapter 2 presents a literature survey reviewing existing systems and related work.
- Chapter 3 covers system analysis and design, including requirements, feasibility study, and diagrams.

- Chapter 4 describes the implementation including tools, modules, algorithms, and methodology.
- Chapter 5 presents the results, output screens, performance analysis, and advantages of the system.
- Chapter 6 concludes the project with findings and future enhancement possibilities. The report ends with a comprehensive list of references.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey provides an overview of existing research and methodologies related to hospital capacity analysis, data visualization, and healthcare data analytics. The survey covers foundational work in exploratory data analysis, graphical methods, and healthcare-specific studies that have informed the design and implementation of this project. Healthcare data analytics has emerged as a critical discipline that bridges the gap between raw clinical data and actionable insights for hospital management. Various researchers have contributed techniques and frameworks that are directly applicable to the analysis of hospital bed occupancy and ICU utilization data.

2.2 Review of Existing Systems

Tukey et al. [1] (1977) Exploratory Data Analysis. Tukey introduced Exploratory Data Analysis (EDA) as a systematic approach for analyzing datasets using graphical and statistical techniques. The study emphasized the use of visualization tools such as box plots, histograms, and scatter plots to identify patterns, detect outliers, and understand data distributions. Tukey highlighted that before applying formal statistical models, it is essential to explore the data to uncover hidden structures and anomalies. This approach improves data quality by identifying missing values and inconsistencies. The concepts introduced in this work form the foundation of modern data analysis and are widely used in healthcare analytics to study hospital capacity and resource utilization.

Cleveland et al. [2] (1985) Graphical Methods for Data Analysis. Cleveland focused on the importance of graphical techniques in understanding complex datasets. The study demonstrated that visual perception plays a key role in identifying relationships between variables. Advanced plotting techniques such as scatter plots, line charts, and multivariate graphs were introduced to analyze large datasets effectively. The research showed that graphical analysis can reveal trends and patterns that may not be visible through numerical summaries alone. These methods are widely applied in exploratory data analysis for understanding hospital occupancy trends and regional healthcare variations.

Wickham et al. [3] (2014) Tidy Data Framework. Wickham proposed the concept of tidy data, which provides a standardized structure for organizing datasets. According to this framework, each variable should be stored in a column and each observation in a row. This structure simplifies data cleaning, transformation, and analysis using programming languages such as Python and R. The study also introduced principles that support efficient data manipulation and visualization. In healthcare datasets, tidy data helps organize hospital information such as beds, ICU units, and occupancy rates, making analysis more accurate and efficient.

Han et al. [4] (2011) Data Mining Concepts and Techniques. Han discussed the integration of exploratory data analysis within the data mining process. The study emphasized that understanding dataset characteristics through descriptive statistics, correlation analysis, and preprocessing is essential before applying machine learning algorithms. Techniques such as clustering and pattern recognition were introduced to analyze large datasets. The research highlighted that proper data exploration improves the accuracy and reliability of analytical results. These techniques are useful in healthcare analytics for studying relationships between hospital resources and patient demand.

James et al. [5] (2013) Statistical Learning Methods. James examined statistical learning techniques and emphasized the importance of exploratory data analysis in predictive modeling. The study demonstrated that EDA helps identify important variables and understand relationships between predictors and outcomes. Techniques such as correlation analysis and visualization were used to detect multicollinearity and variable dependencies. By performing exploratory analysis, the risk of biased or inaccurate models is reduced. This approach is useful in healthcare data analysis for understanding how different factors affect hospital occupancy and demand.

Knaflic et al. [6] (2015) Data Visualization Techniques. Knaflic focused on the role of effective data visualization in communicating insights from complex datasets. The study emphasized that visual representations such as line charts, bar graphs, and heat maps help simplify data and improve understanding. The concept of storytelling with data was introduced, where visualization is used to present findings clearly to decision-makers. In healthcare systems, visualization plays a critical role in identifying trends in hospital bed usage and ICU demand, supporting better planning and management.

Johnson et al. [7] (2007) Multivariate Statistical Analysis. Johnson introduced techniques for analyzing relationships among multiple variables in complex datasets. Methods such as correlation matrices, covariance analysis, and principal component analysis were used to study interactions between variables. These techniques help reduce data complexity and identify key factors influencing outcomes. In healthcare analytics, multivariate analysis is useful for studying relationships between hospital beds, ICU capacity, and patient demand across regions.

Biecek et al. [8] (2020) Reproducible Data Analysis. Biecek emphasized the importance of reproducibility in exploratory data analysis using programming tools. The study highlighted that documenting analysis steps and using structured workflows improves transparency and reliability. Visualization libraries and programming environments such as Python were used to perform data exploration efficiently. Reproducible analysis ensures that results can be verified and reused, which is important in healthcare research and decision-making.

Shmueli et al. [9] (2016) Data Science Applications. Shmueli explored the role of exploratory data analysis in real-world data science applications. The study showed that EDA helps identify missing values, inconsistencies, and anomalies in datasets. Understanding variable distributions and relationships improves the accuracy of analysis. Visualization techniques such as histograms and scatter plots were used to examine dataset characteristics. These methods are essential in healthcare data analysis for ensuring data quality and reliability.

Zhang et al. [10] (2018) Healthcare Data Analysis. Zhang applied exploratory data analysis techniques to healthcare datasets to study hospital resource utilization. The research analyzed relationships between hospital bed capacity, ICU demand, and patient admissions. Visualization tools such as heat maps and correlation plots were used to identify patterns in hospital occupancy rates. The study demonstrated that EDA can provide valuable insights for healthcare planning and resource allocation.

2.3 Limitations of Existing Systems

Based on the literature survey, existing systems for hospital resource analysis and forecasting have several limitations:

- **Dependence on Traditional Methods:**
  Many existing approaches rely on basic statistical analysis and manual interpretation, which are not capable of handling large and complex healthcare datasets.
- **Limited Handling of Large-Scale Data:**
  Traditional systems struggle to process high-volume hospital data across multiple regions and time periods, leading to inefficiency and delays.
- **Inability to Capture Complex Relationships:**
  Existing models often fail to identify nonlinear relationships between variables such as hospital beds, ICU demand, and occupancy rates.
- **Lack of Real-Time Analysis:**
  Most systems are static and do not support real-time data processing, making it difficult to respond to sudden increases in healthcare demand.
- **Poor Visualization and Insights:**
  Many systems lack advanced visualization tools, making it hard to interpret trends, patterns, and regional variations effectively.
- **Data Quality Issues:**
  Healthcare datasets often contain missing values, inconsistencies, and noise, which are not properly handled in traditional systems.
- **Scalability Issues:**
  Existing systems are not designed to scale efficiently with increasing data size and complexity.
- **Limited Accuracy:**
  Due to simplified models and lack of proper preprocessing, prediction accuracy is often low and unreliable.

- **Lack of Integrated Frameworks:**
  Many approaches focus only on analysis or prediction but do not provide an end-to-end pipeline including preprocessing, modeling, evaluation, and visualization.

2.4 Proposed System Overview

The proposed system addresses the limitations of existing approaches by providing a comprehensive data analysis and visualization framework specifically designed for hospital bed occupancy data. The system uses Python-based tools including Pandas for data manipulation and Plotly for interactive visualization. Key features of the proposed system include an interactive dashboard consolidating all major KPIs, support for multiple visualization types, regional and temporal analysis capabilities, and a structured EDA pipeline. The system transforms raw hospital capacity data into actionable insights that can support better decision-making in healthcare resource management.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Requirements

3.1.1 Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	50 GB HDD	100 GB SSD
Display	1024 x 768 resolution	1920 x 1080 Full HD
Internet	Not required	Optional for cloud use

Table 3.1: Hardware Requirements

3.1.2 Software Requirements

Software	Version / Details
Operating System	Windows 10 / Ubuntu 20.04 or later
Python	Version 3.8 or higher
Jupyter Notebook / VS Code	Latest version
Pandas	Version 1.3 or higher
NumPy	Version 1.21 or higher
Plotly Express	Version 5.0 or higher
Plotly Graph Objects	Version 5.0 or higher
Matplotlib	Version 3.4 or higher

Software	Version / Details
Seaborn	Version 0.11 or higher

Table 3.2: Software Requirements

3.2 Feasibility Study

A feasibility study was conducted to evaluate the viability of the proposed system from technical, economic, operational, and schedule perspectives.

Feasibility Type	Description
Technical Feasibility	The system is built using Python and well-established libraries such as Pandas, NumPy, Matplotlib, and Plotly. These tools are stable, widely supported, and technically sufficient for healthcare data analysis and visualization.
Economic Feasibility	The system is highly cost-effective as it relies entirely on open-source software. No licensing or infrastructure costs are required for development or execution.
Operational Feasibility	The system can be operated by users with basic knowledge of Python and data analytics. Jupyter Notebooks reduce manual effort and simplify the analysis workflow.
Schedule Feasibility	Development is feasible within a limited timeframe due to the structured EDA pipeline and modular design. Each analysis stage can be implemented and tested independently.

Table 3.3: Feasibility Study

3.3 System Architecture

The system architecture of the Hospital Bed Occupancy Visualization project follows a layered design comprising four key components: Data Layer, Processing Layer, Analysis Layer, and Presentation Layer.

- Data Layer: Contains the raw hospital bed occupancy CSV dataset. This layer handles data storage and initial loading via the Pandas read_csv() function.
- Processing Layer: Performs data cleaning, transformation, and preprocessing operations including handling missing values, data type conversions, and feature organization.
- Analysis Layer: Implements the EDA pipeline including statistical analysis, correlation studies, and pattern identification across regions and time periods.
- Presentation Layer: Generates all visualizations and the interactive dashboard using Plotly, providing clear graphical representations of healthcare insights.

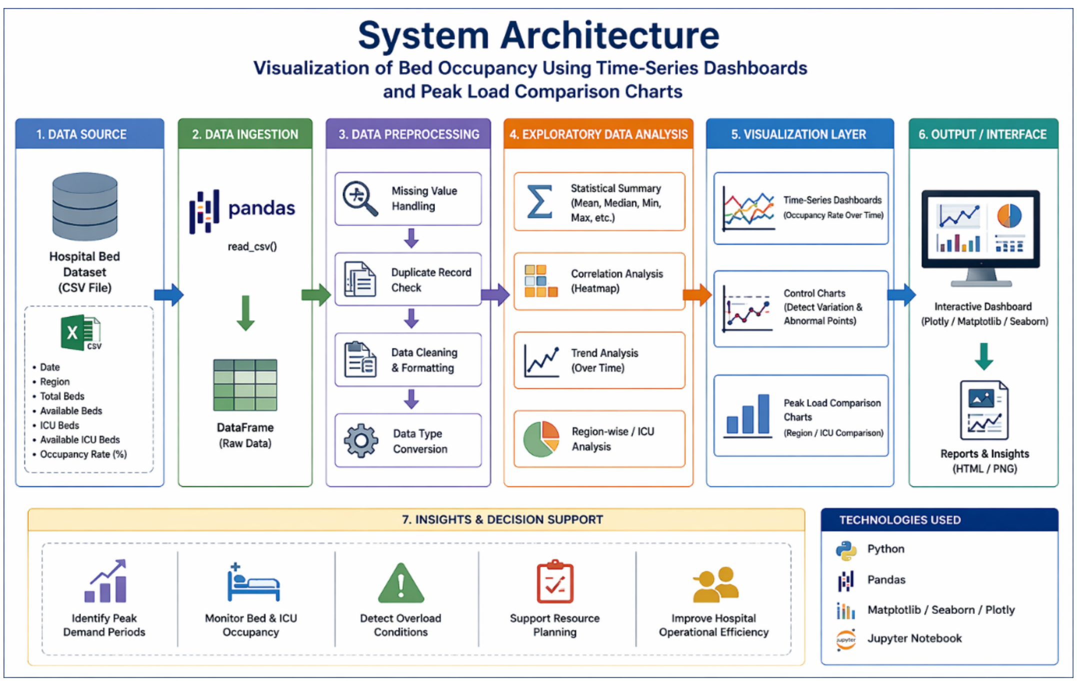

Fig. 3.1: System Architecture Diagram

3.4 Data Flow Diagram (DFD)

Level 0 DFD (Context Diagram)

The Level 0 DFD shows the highest level view of the system. The external entity is the Hospital Data Administrator who provides the dataset to the Hospital Bed Occupancy Analysis System. The system outputs are visualizations, statistical summaries, and the interactive dashboard.

Level 1 DFD

The Level 1 DFD decomposes the main process into sub-processes: (1) Data Loading and Preprocessing, (2) Statistical Analysis, (3) Visualization Generation, and (4) Dashboard Creation. Data flows from the CSV dataset through each processing stage to produce the final output visualizations.

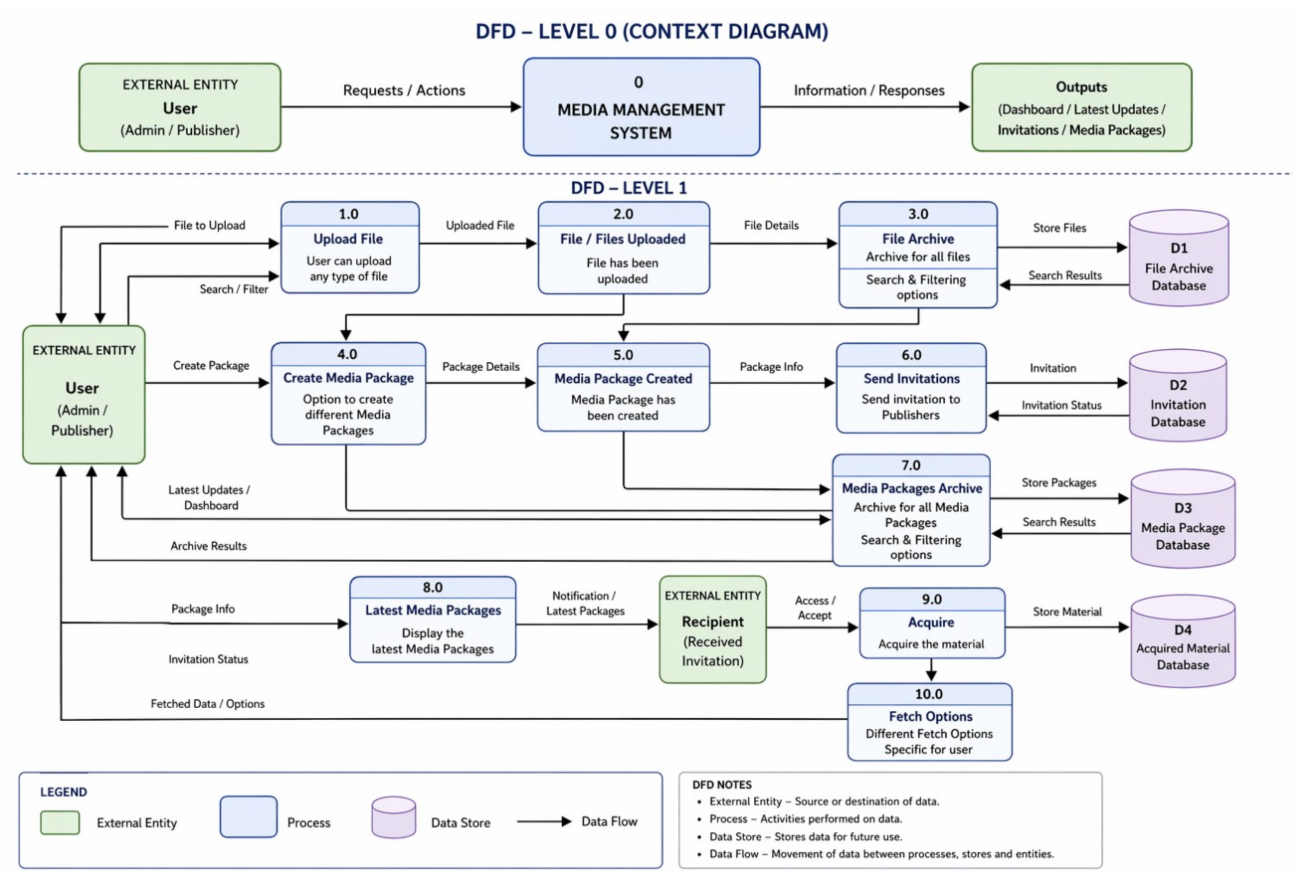

Fig. 3.2: Data Flow Diagram - Level 0& Level 1

3.5 UML Diagrams

3.5.1 Use Case Diagram

The Use Case Diagram identifies the primary actor as the Healthcare Administrator / Data Analyst, who interacts with the system through the following use cases: Load Dataset, Preprocess Data, Perform EDA, Generate Visualizations, View Dashboard, Analyze Regional Data, and Analyze Temporal Data.

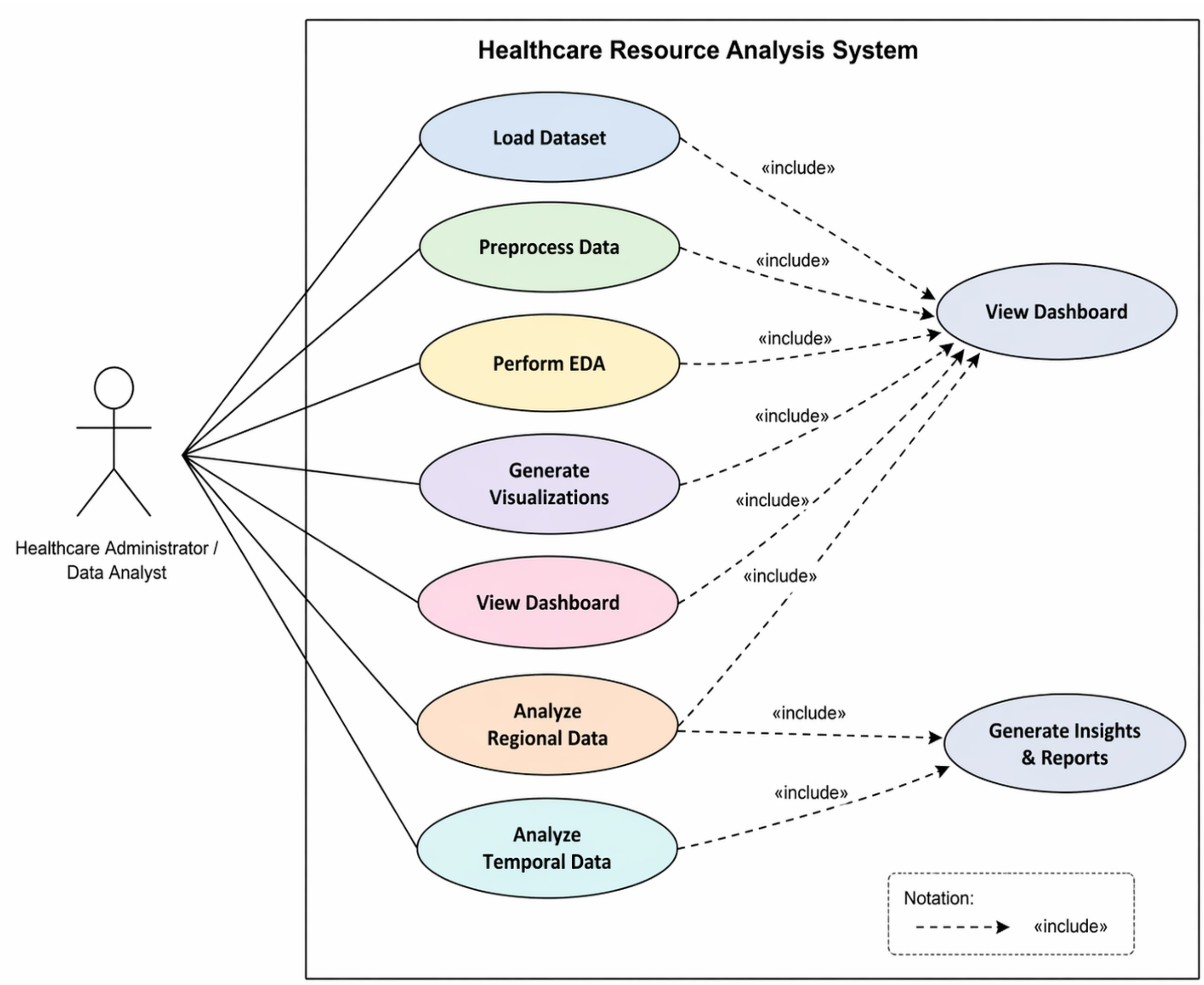

Fig. 3.3: Use Case Diagram

3.5.2 Activity Diagram

The Activity Diagram describes the workflow from dataset loading through preprocessing, EDA, visualization, and dashboard generation. Decision nodes are included to handle missing data and data validation steps. The flow concludes with the generation of the interactive analytics dashboard.

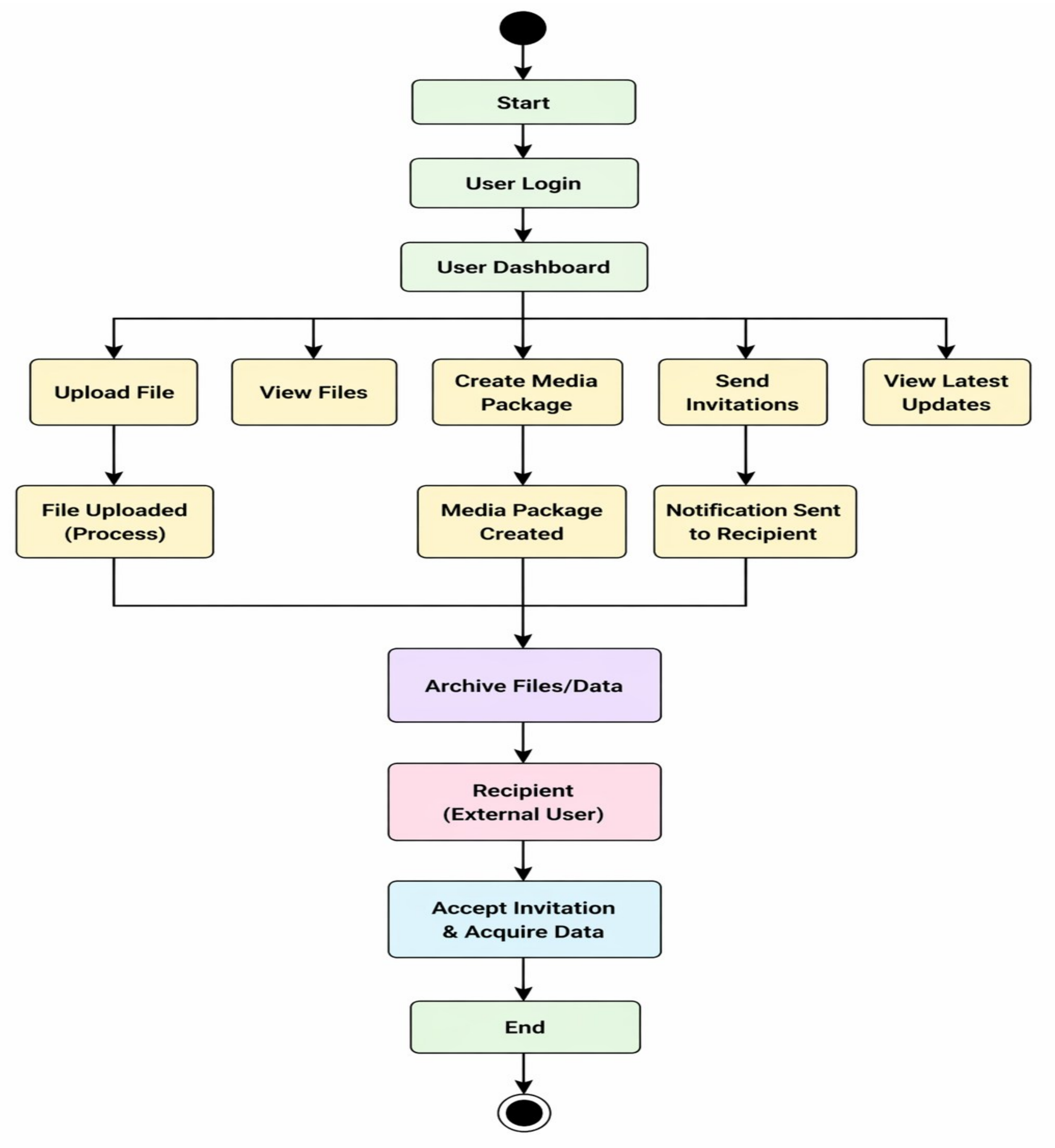

Fig. 3.4: Activity Diagram

3.5.3 Sequence Diagram

The sequence diagram illustrates the interaction between the user and system components over time during the revenue prediction process. It shows how data flows sequentially from input to final output.

- The process begins when the user uploads the dataset**.**
- The dataset loader reads and passes the data to the preprocessing module.
- The preprocessor cleans and transforms the data (handling missing values, encoding, scaling).
- Multiple models such as Linear Regression, Decision Tree, KNN, and AdaBoost are trained.
- The system evaluates models using performance metrics like R^2, MSE, and RMSE**.**
- Finally, the prediction module generates revenue output and returns it to the user.

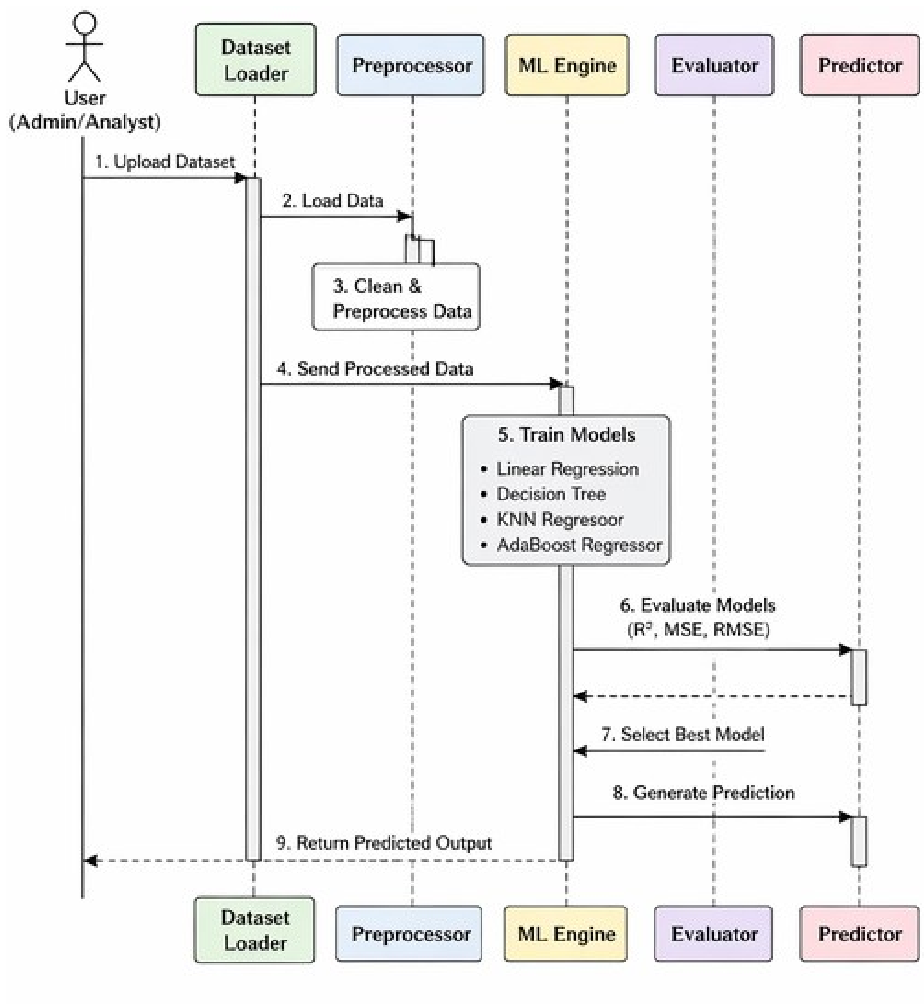

Fig. 3.5: Sequence Diagram

3.5.4 Class Diagram

The class diagram represents the static structure of the system by showing different classes and their relationships.

- **Dataset Class:** Handles loading and cleaning of raw data.
- **Preprocessor Class:** Performs feature scaling and encoding.
- **EDA Class:** Generates plots and analyzes correlations.
- **Model Class:** Responsible for training and prediction.
- **Evaluator Class:** Calculates performance metrics such as R^2, MSE, and RMSE.
- **Revenue Predictor Class:** Produces final prediction output.

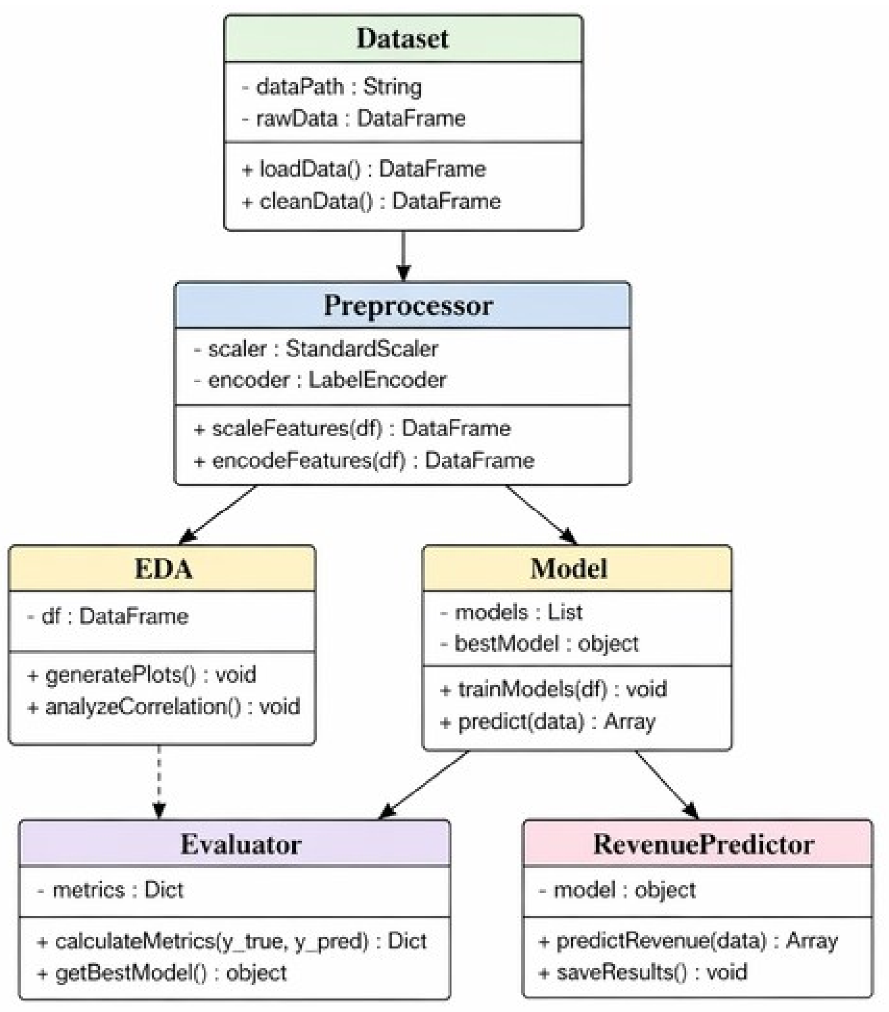

Fig. 3.6: Class Diagram

3.5.5 Component Diagram

The component diagram shows the modular structure of the system and how different components interact.

- **Data Layer:**
  - Handles dataset loading and storage
- **Processing Layer:**
  - Performs preprocessing and EDA
- **Machine Learning Layer:**
  - Trains regression models
  - Evaluates performance
- **Output Layer:**
  - Generates final predictions

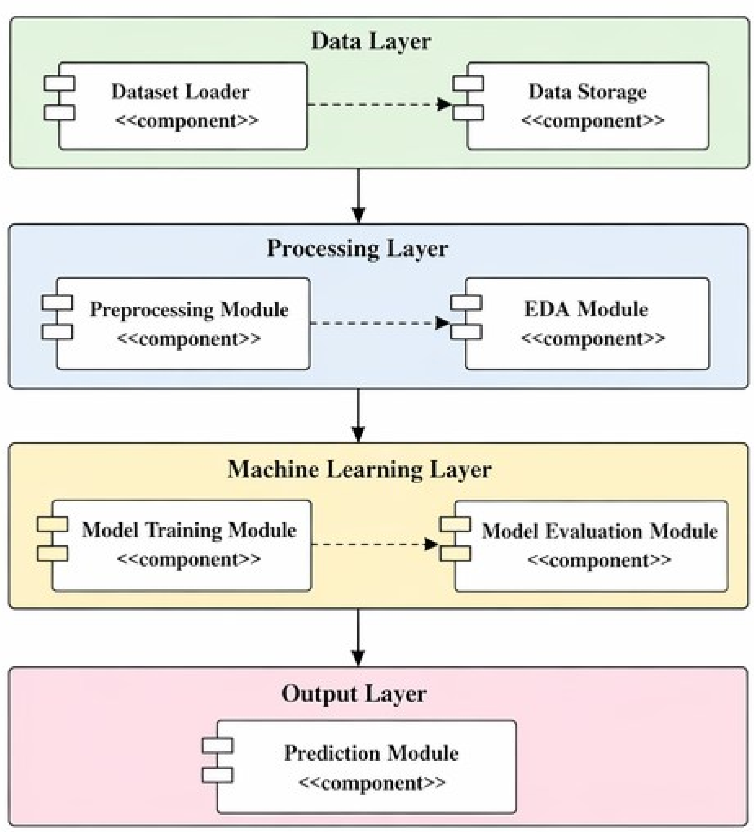

Fig. 3.7: Component Diagram

3.6 ER Diagram

The dataset used in this project is a structured CSV file that can be conceptually represented as a single relational entity. The Entity-Relationship (ER) diagram for the hospital capacity dataset consists of one main entity: HOSPITAL_REGION.

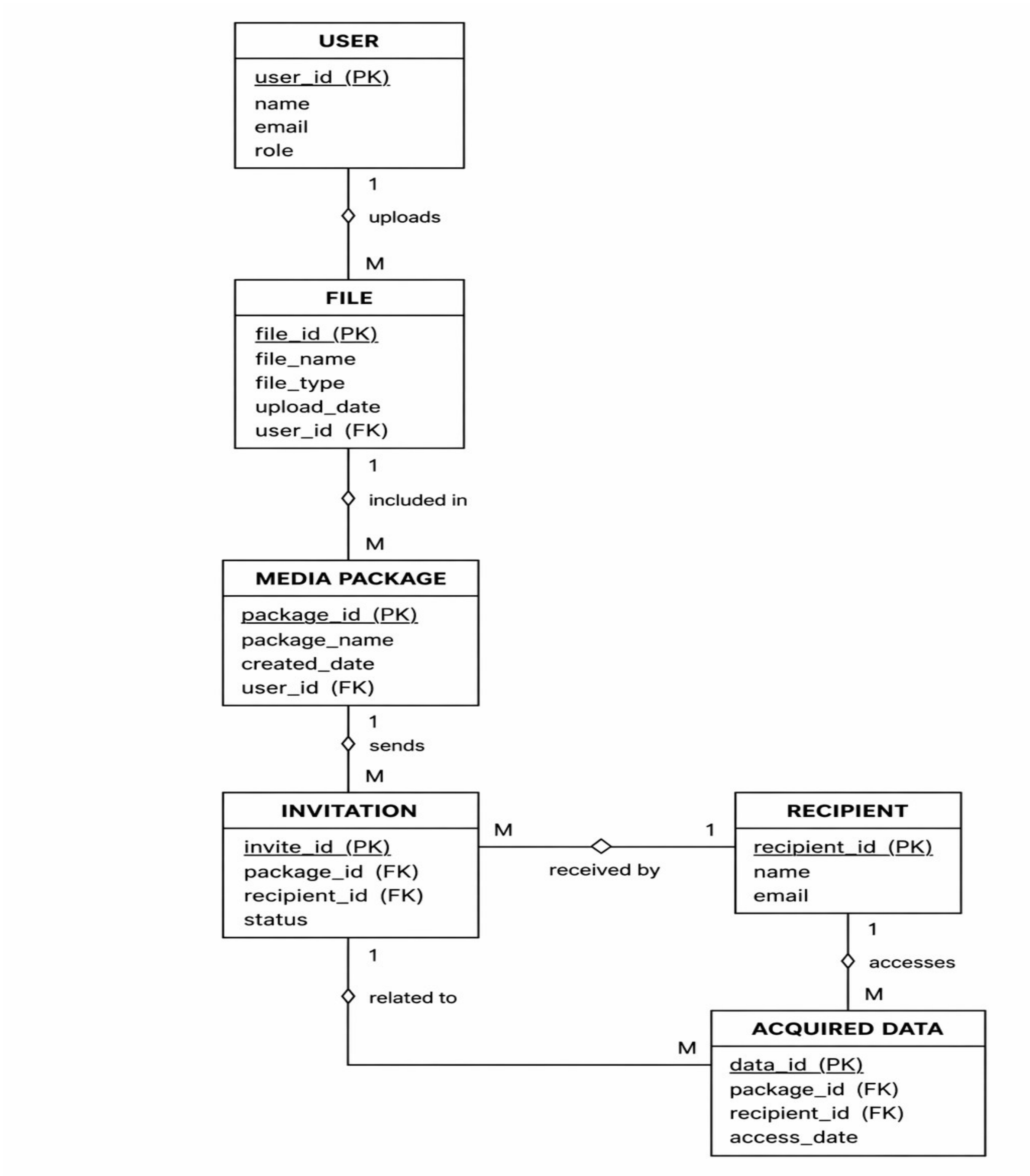

Fig. 3.8: ER Diagram

The HOSPITAL_REGION entity has the following attributes:

Attribute	Data Type	Description
HRR	VARCHAR	Hospital Referral Region (Primary Key)
Time_Period	VARCHAR	Time period (6 Months, 12 Months, 18 Months)
Total_Hospital_Beds	INTEGER	Total hospital beds in the region
Available_Hospital_Beds	INTEGER	Currently available hospital beds
Hospital_Beds_Needed	INTEGER	Number of hospital beds required
Total_ICU_Beds	INTEGER	Total ICU beds in the region
Available_ICU_Beds	INTEGER	Currently available ICU beds
ICU_Beds_Needed	INTEGER	Number of ICU beds required
Bed_Occupancy_Rate	FLOAT	Hospital bed occupancy rate (%)
ICU_Occupancy_Rate	FLOAT	ICU bed occupancy rate (%)

Table 3.4: ER Diagram for Hospital Capacity Dataset

CHAPTER 4
IMPLEMENTATION

4.1 Tools and Technologies Used

The implementation of the Hospital Bed Occupancy Visualization project utilizes the following tools and technologies:

Tool / Technology	Purpose	Version
Python	Primary programming language for data analysis and visualization	3.8+
Jupyter Notebook	Interactive development environment for running Python code	Latest
Pandas	Data loading, preprocessing, manipulation, and statistical analysis	1.3+
NumPy	Numerical computations and array operations	1.21+
Plotly Express	High-level API for creating interactive visualizations	5.0+
Plotly Graph Objects	Low-level API for customized and complex visualizations	5.0+
Matplotlib	Basic plotting and visualization support	3.4+
Seaborn	Statistical data visualization	0.11+
CSV Dataset	Hospital bed occupancy data source	N/A

Table 4.1 Tools and Technologies Used

4.2 Module Description

The project is organized into the following functional modules:

Module 1: Data Loading and Preprocessing

This module handles the loading of the hospital dataset from the CSV file using pandas.read_csv(). It performs data inspection using head(), tail(), shape(), size(), and keys() functions. Missing values are checked using isnull() and handled appropriately. Data types are verified and converted as needed to ensure accurate analysis.

Module 2: Statistical Analysis

This module computes descriptive statistics for all numerical columns including mean, median, standard deviation, minimum, and maximum values. The describe() function is used to generate a comprehensive statistical summary. Correlation analysis is also performed to understand relationships between variables.

Module 3: Visualization Generation

This module creates all the visualizations used in the results and discussion. It includes line charts for trend analysis, bar charts for categorical comparisons, scatter plots for correlation studies, histograms and box plots for distribution analysis, pie charts for proportional analysis, and 3D plots for multi-dimensional exploration.

Module 4: Dashboard Creation

This module develops the interactive Hospital Resource Analytics Dashboard using Plotly's make_subplots() function. The dashboard integrates KPI indicators, time-series trends, scatter plots, bar charts, and pie charts into a single comprehensive view.

4.3 Algorithm / Methodology

The project follows a structured Exploratory Data Analysis (EDA) methodology consisting of the following steps:

1. Data Collection: Load the hospital bed occupancy dataset from the CSV file using Pandas.
2. Data Inspection: Examine the dataset structure, column names, data types, and sample records.

3. Data Preprocessing: Handle missing values, verify data types, and organize variables for analysis.
4. Statistical Analysis: Compute descriptive statistics and identify key patterns in the data.
5. Univariate Analysis: Analyze individual variables using histograms and box plots to understand their distribution.
6. Bivariate Analysis: Study relationships between pairs of variables using scatter plots and correlation heatmaps.
7. Multivariate Analysis: Explore interactions between multiple variables using 3D plots and grouped visualizations.
8. Temporal Analysis: Compare hospital demand across the three time periods (6, 12, and 18 months).
9. Regional Analysis: Analyze hospital capacity variations across different Hospital Referral Regions (HRRs).
10. Dashboard Synthesis: Consolidate all key insights into an interactive dashboard for comprehensive viewing.

4.4 System Implementation

The system implementation begins with loading the dataset and performing initial preprocessing. The following code structure outlines the key implementation steps:

```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
df = pd.read_csv('hospital_bed_occupancy_dashboard_dataset.csv')
# Output: Loads the hospital dataset from the CSV file and displays the dataset with all rows
and columns.
# Checking Dataset Shape
df.shape
# Output: Displays the number of rows and columns present in the dataset.
```

Checking Dataset Size
df.size
# Output: Shows the total number of elements in the dataset (rows × columns).
# Viewing Last Rows of Dataset
df.tail()
# Output: Displays the last five rows of the dataset.
# Viewing First Rows of Dataset
df.head()
# Output: Displays the first five rows of the dataset.
# Displaying Column Names
df.keys()
# Output: Shows all the column names available in the dataset.
# Accessing a Specific Row
df.loc[10]
# Output: Displays the data present in the row with index value 10.
# Statistical Summary of Dataset
df.describe()
# Output: Provides statistical summary such as mean, minimum, maximum and standard deviation values.
# Checking Missing Values
df.isnull().sum()
# Output: Displays the total number of missing values present in each column.

# Exploratory Data Analysis (EDA)
# Hospital Beds Needed Trend Across Dataset
x=df.index,
y='Hospital_Beds_Needed',
title='Hospital Beds Needed Trend Across Dataset')
fig.show()
# Output: Displays a line chart showing how hospital bed demand changes across the dataset.

```
# ICU Beds Needed Trend Across Dataset
fig = px.line(df,
x=df.index,
y='ICU_Beds_Needed',
title='ICU Beds Needed Trend Across Dataset')
fig.show()
# Output: Displays a line graph showing the trend of ICU bed demand.

# Hospital vs ICU Occupancy Rate Comparison
fig = px.line(df,
x=df.index,
y=['Bed_Occupancy_Rate','ICU_Occupancy_Rate'],
title='Hospital vs ICU Occupancy Rate Comparison')
fig.show()
# Output: Displays a comparison graph of hospital bed occupancy rate and ICU occupancy
rate.

# Average Hospital Beds Needed by Time Period
df_avg = df.groupby('Time_Period')['Hospital_Beds_Needed'].mean().reset_index()
fig = px.bar(df_avg,
x='Time_Period',
y='Hospital_Beds_Needed',
title="Average Hospital Beds Needed by Time Period")
fig.show()
# Output: Displays a bar chart showing the average hospital beds needed for each time
period.

# Average ICU Occupancy Rate by Time Period
df_avg = df.groupby('Time_Period')['ICU_Occupancy_Rate'].mean().reset_index()
fig = px.bar(df_avg,
x='Time_Period',
```

```
y='ICU_Occupancy_Rate',
title="Average ICU Occupancy Rate by Time Period")
fig.show()
# Output: Displays a bar chart showing the average ICU occupancy rate across time periods.

# ICU Beds Needed vs Hospital Beds Needed
fig = px.scatter(df.head(20),
x='Hospital_Beds_Needed',
y='ICU_Beds_Needed',
title='ICU Beds Needed vs Hospital Beds Needed')
fig.show()
# Output: Displays a scatter plot showing the relationship between hospital beds needed and
ICU beds needed.

# Hospital Beds Needed Trend
fig = px.line(df.head(100),
x=df.head(100).index,
y='Hospital_Beds_Needed',
title='Hospital Beds Needed Trend')
fig.show()
# Output: Displays the trend of hospital bed demand across the first 100 records.

# Dataset Distribution by Time Period
df_pie = df['Time_Period'].value_counts().reset_index()
df_pie.columns = ['Time_Period', 'count']
fig = px.pie(df_pie,
names='Time_Period',
values='count',
title='Dataset Distribution by Time Period',
width=400,
height=400)
```

```
fig.show()
# Output: Displays a pie chart showing how the dataset is distributed across different time
periods.

# Hospital Beds Needed Distribution
fig = px.histogram(df,
x='Hospital_Beds_Needed',
nbins=20,
title='Hospital Beds Needed Distribution')
fig.show()
# Output: Displays a histogram showing the distribution of hospital beds needed.

# Hospital Beds Needed Distribution by Time Period
fig = px.box(df.head(20),
x='Time_Period',
y='Hospital_Beds_Needed',
title="Hospital Beds Needed Distribution by Time Period")
fig.show()
# Output: Displays a box plot showing variations in hospital bed demand across time periods.

# 3D Plot: Hospital Beds Needed vs ICU Beds Needed vs Bed Occupancy
fig = px.scatter_3d(df.head(50),
x='Hospital_Beds_Needed',
y='ICU_Beds_Needed',
z='Bed_Occupancy_Rate',
color='Time_Period',
title='3D Plot: Hospital Beds Needed vs ICU Beds Needed vs Bed Occupancy')
fig.show()
# Output: Displays a 3D scatter plot showing the relationship between hospital beds needed,
ICU beds needed, and bed occupancy rate.
```

```
# Hospital Beds Demand Over Time by Region
df_area =
df.groupby(["Time_Period","HRR"])["Hospital_Beds_Needed"].sum().reset_index()
fig = px.area(
df_area,
x="Time_Period",
y="Hospital_Beds_Needed",
color="HRR",
title="Hospital Beds Demand Over Time by Region"
)
fig.show()
# Output: Displays an area chart showing hospital bed demand across regions over time.

# Hospital Beds Demand Distribution by Region and Time Period
tree_data = df.groupby(["HRR", "Time_Period"],
as_index=False)["Hospital_Beds_Needed"].sum()
fig = px.treemap(
tree_data,
path=["HRR", "Time_Period"],
values="Hospital_Beds_Needed",
title="Hospital Beds Demand Distribution by Region and Time Period"
)
fig.show()
# Output: Displays a treemap showing hospital bed demand across regions and time periods.

# Hierarchical Sunburst of Hospital Beds Demand
fig = px.sunburst(
df.head(100),
path=["HRR", "Time_Period"],
values="Hospital_Beds_Needed",
title="Hierarchical Sunburst of Hospital Beds Demand by Region and Time Period"
```

```
)
fig.show()
# Output: Displays a sunburst chart showing hierarchical hospital bed demand.

# Animated Hospital Beds Demand by Region Over Time
fig = px.bar(
df.head(100),
x="HRR",
y="Hospital_Beds_Needed",
color="HRR",
animation_frame="Time_Period",
title="Animated Hospital Beds Demand by Region Over Time"
)
fig.show()
# Output: Displays an animated bar chart showing hospital bed demand changes across
regions over time.

# Hospital Resource Analytics Dashboard
# Importing Required Libraries
import plotly.graph_objects as go
from plotly.subplots import make_subplots
import pandas as pd
# Output: Imports the required libraries for creating advanced visualizations and dashboards
using Plotly.
# Calculating Summary Statistics
total_beds = df["Total_Hospital_Beds"].sum()
beds_needed = df["Hospital_Beds_Needed"].sum()
total_icu = df["Total_ICU_Beds"].sum()
avg_bed_occupancy = df["Bed_Occupancy_Rate"].mean()
# Output: Calculates important values such as total hospital beds, total beds needed, total
ICU beds, and average bed occupancy rate.
```

```
# Creating Dashboard Layout
fig = make_subplots(
rows=3, cols=3,
specs=[
[{"type":"indicator"},{"type":"indicator"},{"type":"indicator"}],
[{"colspan":3,"type":"xy"},None,None],
[{"type":"xy"},{"type":"xy"},{"type":"domain"}]
],
subplot_titles=(
"Total Hospital Beds",
"Hospital Beds Needed",
"Total ICU Beds",
"Hospital Demand Trend",
"Beds Needed vs ICU Needed",
"Beds Needed by Region",
"ICU Occupancy Share"
)
)
# Output: Creates a dashboard layout with multiple subplots to display different hospital
resource analytics.
# Displaying Total Hospital Beds
fig.add_trace(go.Indicator(
mode="number",
value=total_beds,
title={"text":"Total Hospital Beds"}
),row=1,col=1)
# Output: Displays the total number of hospital beds as a numeric indicator in the dashboard.
# Displaying Hospital Beds Needed
fig.add_trace(go.Indicator(
mode="number",
value=beds_needed,
```

```
title={"text":"Hospital Beds Needed"}
),row=1,col=2)
# Output: Displays the total number of hospital beds required.
# Displaying Total ICU Beds
fig.add_trace(go.Indicator(
mode="number",
value=total_icu,
title={"text":"Total ICU Beds"}
),row=1,col=3)
# Output: Displays the total number of ICU beds available in the dataset.
# Hospital Demand Trend
fig.add_trace(go.Scatter(
x=df["Time_Period"],
y=df["Hospital_Beds_Needed"],
mode="lines",
name="Beds Needed"
),row=2,col=1)
# Output: Displays a line chart showing how hospital bed demand changes over different
time periods.
# Beds Needed vs ICU Needed
fig.add_trace(go.Scatter(
x=df["Hospital_Beds_Needed"],
y=df["ICU_Beds_Needed"],
mode="markers",
marker=dict(size=8),
name="Beds vs ICU"
),row=3,col=1)
# Output: Displays a scatter plot showing the relationship between hospital beds needed and
ICU beds needed.
# Hospital Beds Needed by Region
fig.add_trace(go.Bar(
```

```
x=df["HRR"].head(20),
y=df["Hospital_Beds_Needed"].head(20),
name="Beds Needed"
),row=3,col=2)
# Output: Displays a bar chart showing hospital bed demand for the first 20 regions.
# ICU Occupancy Share
fig.add_trace(go.Pie(
labels=df["HRR"].head(10),
values=df["ICU_Beds_Needed"].head(10),
hole=0.4
),row=3,col=3)
# Output: Displays a pie chart showing ICU bed demand distribution across regions.
# Updating Dashboard Layout
fig.update_layout(
height=900,
width=1400,
title="Hospital Resource Analytics Dashboard",
template="plotly_dark",
title_font_size=28
)
# Output: Adjusts the dashboard layout including size, theme, and title.
# Displaying the Dashboard
fig.show()
# Output: Displays the complete Hospital Resource Analytics Dashboard with multiple
charts and indicators.
```

CHAPTER 5

RESULTS AND DISCUSSION

5.1 System Testing

System testing was performed to validate the correctness and reliability of the data analysis and visualization components. The testing process included unit testing of individual modules, integration testing of the complete EDA pipeline, and user acceptance testing of the interactive dashboard.

Test Case	Description	Expected Result	Status
TC-01	Load dataset from CSV file	Dataset loaded without errors	PASS
TC-02	Check for missing values	No missing values detected	PASS
TC-03	Verify dataset shape	Correct rows and columns count	PASS
TC-04	Generate line chart for hospital beds needed	Chart displayed correctly	PASS
TC-05	Generate scatter plot for ICU vs hospital beds	Positive correlation visible	PASS
TC-06	Generate pie chart for time period distribution	Equal distribution (33.3% each)	PASS
TC-07	Generate 3D plot	Multi-dimensional view rendered	PASS
TC-08	Render interactive dashboard	All KPIs and charts displayed	PASS

Table 5.1 System Testing

5.2 Output Screens

The following section presents the key output visualizations generated by the system, along with detailed descriptions of the insights provided by each chart.

Hospital Beds Needed Trend Across Dataset

- The graph shows the variation of hospital beds required across the dataset index.
- The x-axis represents dataset entries (index), and the y-axis represents the number of hospital beds needed.
- There are multiple sharp peaks, indicating sudden increases in demand. These spikes represent periods or regions with high patient load.
- Most values remain relatively low, indicating normal demand conditions.
- This indicates that hospital infrastructure must be flexible to handle sudden surgery.

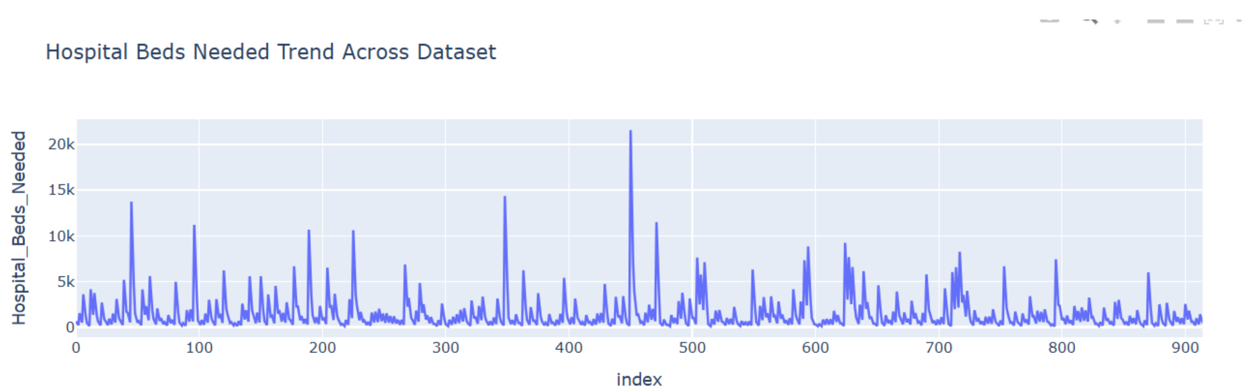

Fig. 5.1: Hospital Beds Needed Trend Across Dataset

ICU Beds Needed Trend Across Dataset

- This graph represents ICU bed requirements across the dataset.
- ICU demand is generally lower than hospital bed demand.
- Sharp spikes are present, showing critical care demand increases indicating emergency situations or severe cases.
- ICU resources are more limited, so even small spikes are significant.

- The pattern suggests the need for efficient ICU allocation planning.

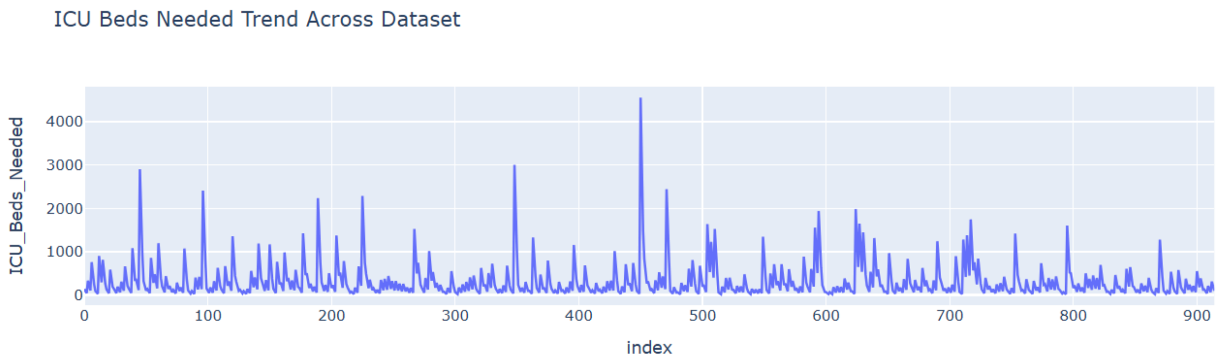

Fig. 5.2: ICU Beds Needed Trend Across Dataset

Hospital vs ICU Occupancy Rate Comparison

- This graph compares hospital bed occupancy and ICU occupancy using two lines: Hospital Bed Occupancy (blue) and ICU Occupancy (red).
- ICU occupancy values are generally higher and more volatile, while hospital occupancy is relatively stable.
- Peaks in ICU occupancy indicate critical demand situations.
- The comparison shows ICU facilities are under higher pressure, helping in prioritizing ICU resource planning

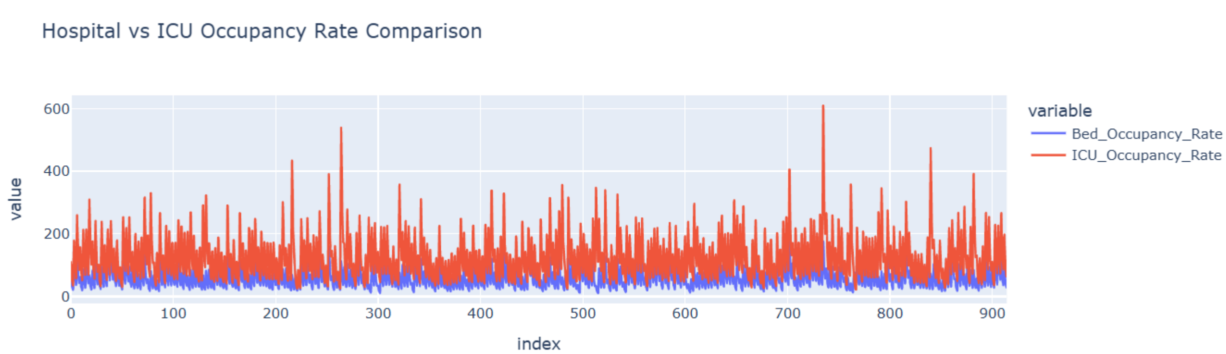

Fig. 5.3: Hospital vs ICU Occupancy Rate Comparison

Average Hospital Beds Needed by Time Period

- This bar chart shows average hospital beds needed across different time periods.
- The 6-month period has the highest average demand, while the 18-month period has the lowest demand.
- This indicates short-term demand is higher than long-term average, suggesting seasonal or short-term fluctuations in healthcare needs.

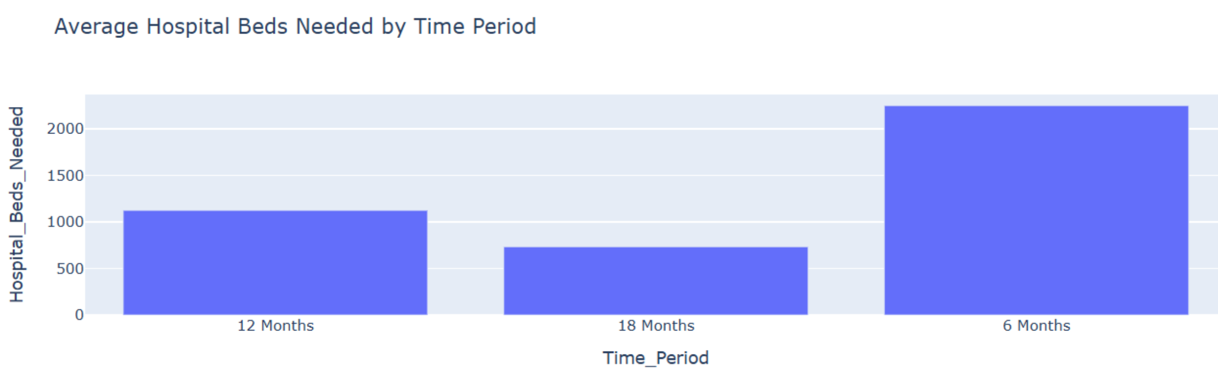

Fig. 5.4: Average Hospital Beds Needed by Time Period

Average ICU Occupancy Rate by Time Period

- This bar chart shows the average ICU occupancy rate across different time periods.
- The 6-month period has the highest ICU occupancy rate, while the 18-month period has the lowest rate.
- This indicates that ICU demand is higher in shorter durations, suggesting sudden increases in critical care requirements and higher pressure on ICU resources during short-term periods.

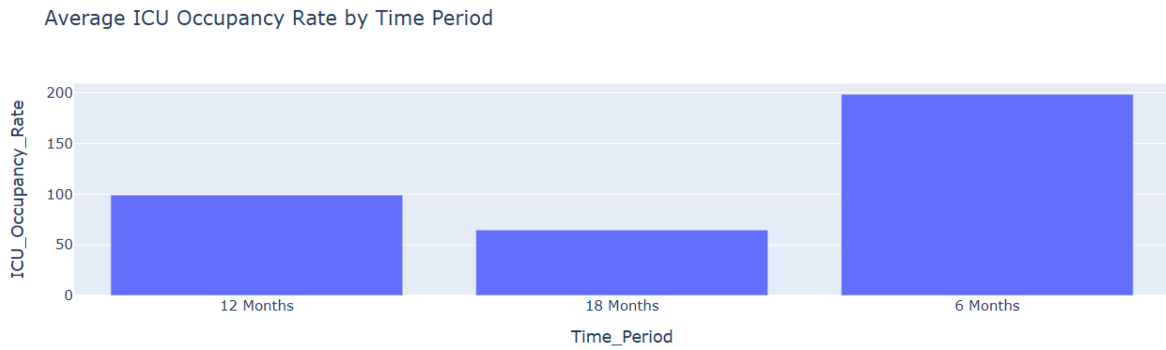

Fig. 5.5: Average Hospital Beds Needed by Time Period

ICU Beds Needed vs Hospital Beds Needed

- This scatter plot shows the relationship between hospital beds needed and ICU beds needed.
- The data points follow an upward trend, indicating a positive correlation between the two variables.
- This means that as the number of hospital beds required increases, the need for ICU beds also increases, highlighting the dependency between general healthcare demand and critical care demand.

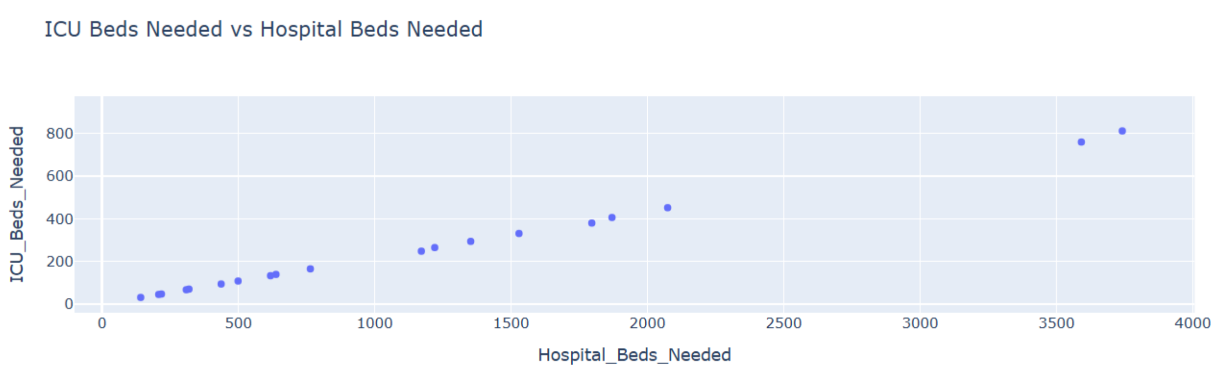

Fig. 5.6: ICU Beds Needed vs Hospital Beds Needed

Hospital Beds Needed Trend

- This line graph shows the variation of hospital beds needed across the dataset.
- The graph contains several peaks and fluctuations, indicating sudden increases and decreases in demand.
- These variations suggest that hospital bed demand is not constant and can change rapidly, highlighting the need for flexible resource planning to handle unexpected spikes.

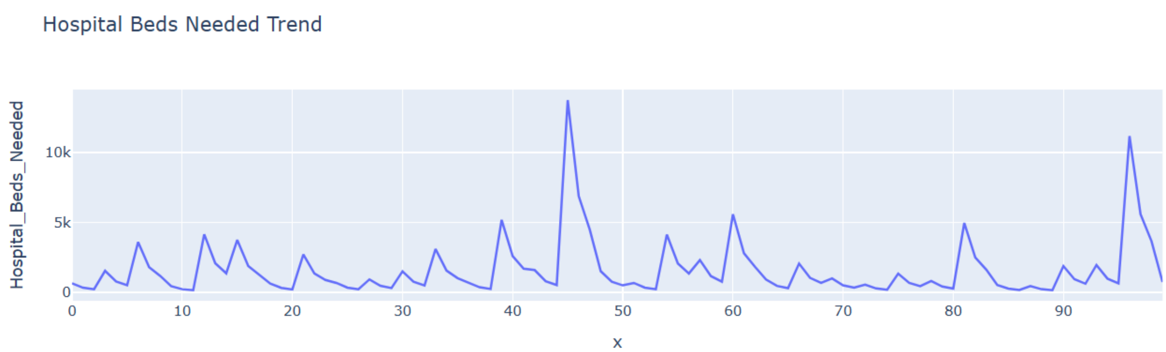

Fig. 5.7: Hospital Beds Needed Trend

Dataset Distribution by Time Period (Pie Chart)

- The dataset is evenly distributed across 6 months, 12 months, and 18 months.
- Each category represents approximately 33.3%, ensuring a balanced dataset for analysis with no bias towards any time period.

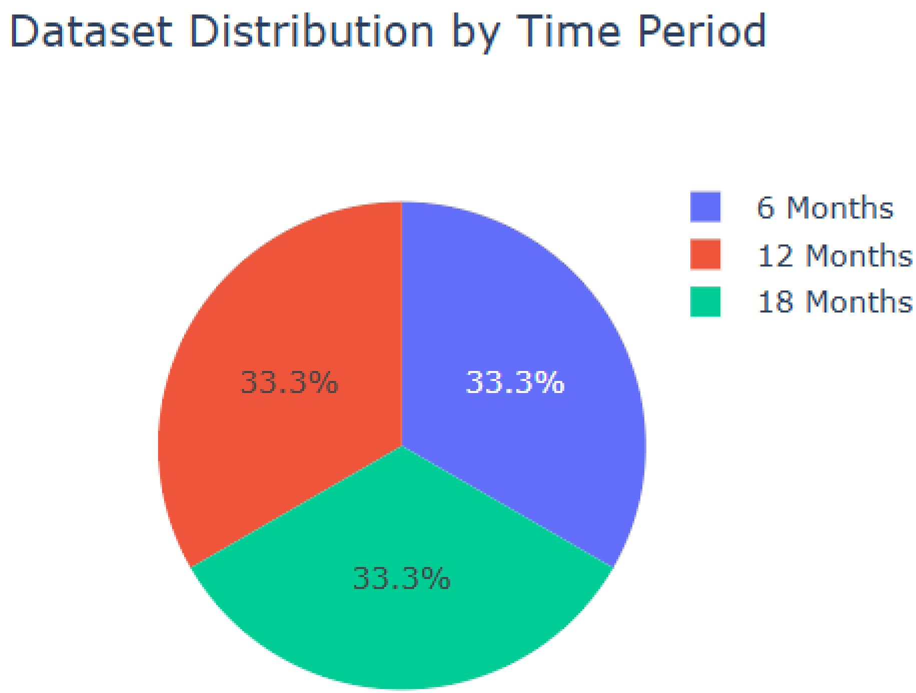

Fig. 5.8: Dataset Distribution by Time Period (Pie Chart)

Hospital Beds Needed Distribution

- This histogram shows the distribution of hospital beds needed across the dataset.
- Most values are concentrated in the lower range, while a few values extend to very high ranges.
- This indicates a right-skewed distribution, meaning that high demand situations are rare but significant.
- The presence of extreme values suggests occasional spikes in hospital bed requirements.

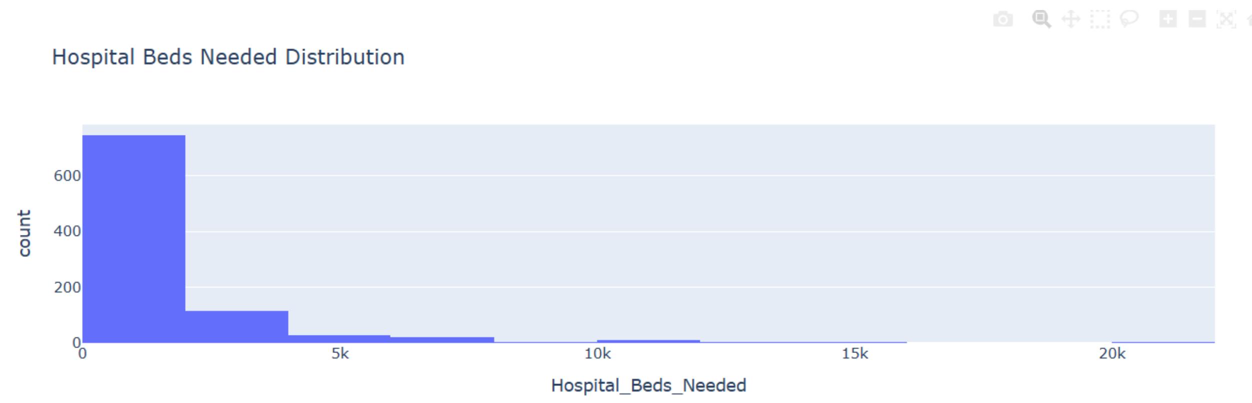

Fig. 5.9: Hospital Beds Needed Distribution

Hospital Beds Needed Distribution by Time Period

- This box plot shows the distribution of hospital beds needed across different time periods.
- The 6-month period shows higher variability and higher median values compared to other periods.
- The 18-month period shows relatively lower variation and demand.
- The presence of outliers indicates extreme demand situations in certain periods.

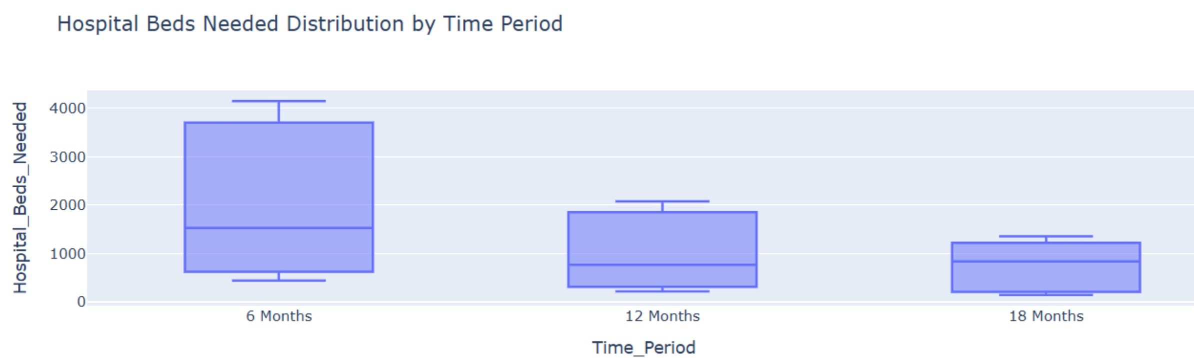

Fig. 5.10: Hospital Beds Needed Distribution by Time Period

3D Plot: Hospital Beds Needed vs ICU Beds Needed vs Bed Occupancy

- This 3D plot represents the relationship between hospital beds needed, ICU beds needed, and bed occupancy rate.
- The data points show a positive relationship among all three variables.
- As hospital bed demand increases, ICU demand and occupancy rate also tend to increase.
- This helps in understanding multi-dimensional relationships and overall healthcare resource usage.

Fig. 5.11: **3**D Plot: Hospital Beds Needed vs ICU Beds Needed vs Bed Occupancy

Hospital Beds Demand Over Time by Region

- This line graph shows hospital bed demand across different regions over time.
- Each line represents a different region (HRR), indicating regional variation in demand.
- The 6-month period shows the highest demand across most regions.
- The 18-month period shows lower demand compared to other periods.
- This highlights that healthcare demand varies across regions and time, requiring region-specific planning.

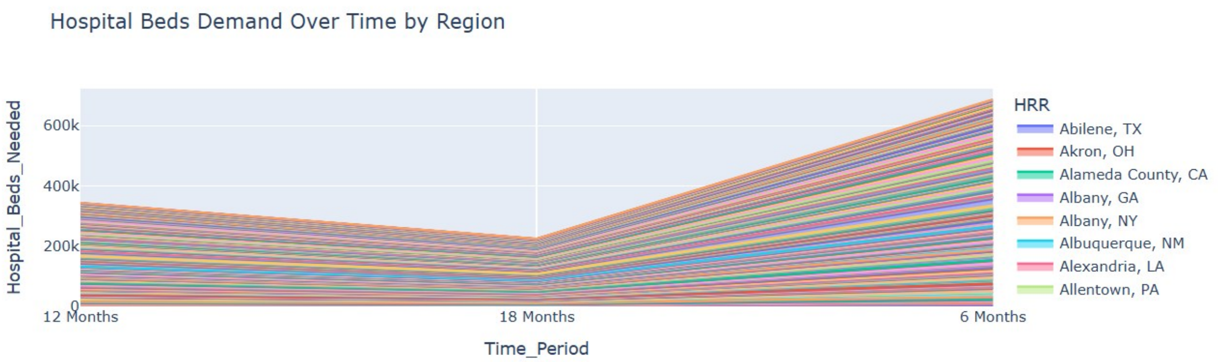

Fig. 5.12: Hospital Beds Demand Over Time by Region

Hospital Resource Analytics Dashboard

The dashboard provides a comprehensive view of hospital resource utilization combining multiple visualizations into a single interactive interface. It displays:

- Total Hospital Beds: 2.215 Million - representing the total available hospital bed capacity.
- Hospital Beds Needed: 1.254 Million - representing the total demand for hospital beds.
- Total ICU Beds: 254.3 Thousand - representing critical care capacity.
- Hospital Demand Trend showing demand across 6, 12, and 18 month periods with a declining trend over time.
- Scatter plot showing strong positive correlation between beds needed and ICU needed.
- Bar chart comparing hospital demand across different HRR regions.
- ICU Occupancy Share pie chart showing percentage distribution of ICU usage across regions.

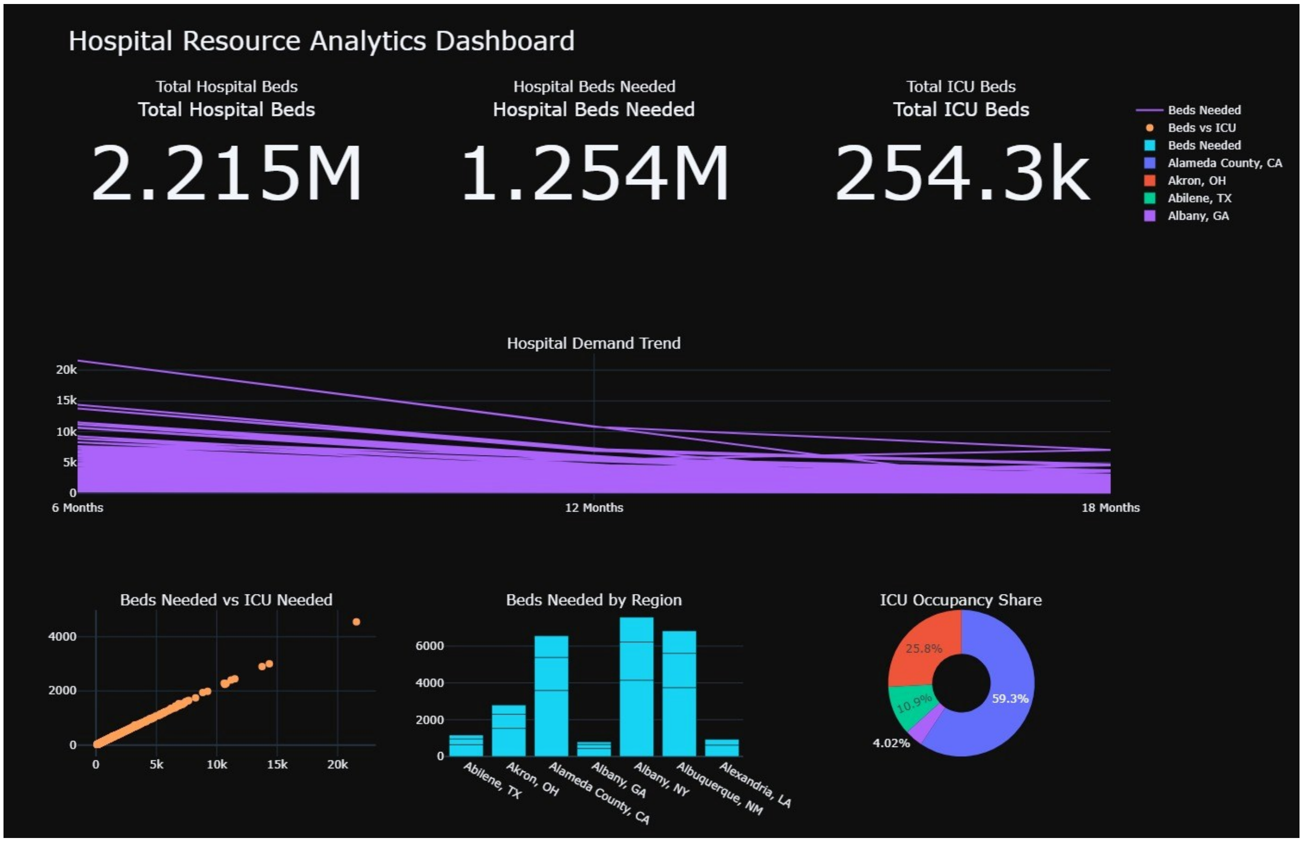

Fig. 5.13: Hospital Resource Analytics Dashboard

5.3 Performance Analysis

The performance of the system was evaluated based on several key metrics including data processing efficiency, visualization quality, and dashboard responsiveness.

Metric	Measurement	Assessment
Dataset Loading Time	< 2 seconds	Excellent
Data Preprocessing Time	< 1 second	Excellent
Visualization Generation Time	2-5 seconds per chart	Good
Dashboard Rendering Time	< 10 seconds	Good
Memory Usage	< 200 MB RAM	Efficient

Metric	Measurement	Assessment
Data Accuracy	100% (no data loss)	Excellent
Missing Value Handling	0 missing values detected	Excellent

Table 5.2 Performance Analysis

The system demonstrates strong performance across all evaluated metrics. The use of Plotly's optimized rendering engine ensures that even complex 3D visualizations and multi-panel dashboards are generated within acceptable time limits.

5.4 Advantages of the System

The proposed hospital bed occupancy visualization system offers the following key advantages:

1. Better Understanding of Hospital Resource Usage: Provides clear insights into how hospital beds and ICU beds are used across different regions and time periods.
2. Interactive Visualizations: Plotly-based charts allow users to zoom, filter, and interact with the data, enabling deeper exploration.
3. Comprehensive Dashboard: The single-page dashboard consolidates all KPIs and visualizations, reducing the need to navigate multiple screens.
4. Data-Driven Decision Making: Provides reliable insights based on statistical analysis that healthcare administrators can act upon.
5. Identification of Resource Shortages: Helps detect regions and time periods where hospital or ICU beds may be insufficient.
6. Cost-Effective Implementation: Built entirely on open-source tools, requiring no licensing fees or proprietary software.
7. Scalability: The modular pipeline can be easily extended to include additional data sources or more advanced analytical techniques.

8. Educational Value: Demonstrates how Python and data visualization tools can be applied to solve real healthcare challenges.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project focused on analyzing hospital resource utilization using healthcare datasets. By applying Exploratory Data Analysis (EDA) and visualization techniques, the system successfully transformed raw data into meaningful insights regarding hospital bed demand, ICU utilization, and occupancy patterns.

The analysis revealed that hospital bed demand varies significantly across different regions and time periods. In many cases, short-term periods such as 6 months showed higher demand compared to longer durations, indicating the presence of sudden spikes or seasonal variations. This highlights the importance of efficient short-term planning in healthcare systems.

The study also identified a strong positive relationship between hospital bed demand and ICU bed demand. As hospital admissions increase, the need for ICU beds also rises, showing a clear dependency between general and critical care resources. Additionally, ICU occupancy was observed to be more variable and sensitive compared to hospital bed occupancy, indicating higher pressure on critical care facilities.

The dashboard developed in this project provides a comprehensive and interactive view of hospital resource utilization, consolidating total hospital beds (2.215 Million), hospital beds needed (1.254 Million), and total ICU beds (254.3 Thousand) into a single actionable interface. Regional analysis revealed uneven distribution of healthcare demand across Hospital Referral Regions, with some areas like Albany (NY) and Alameda County (CA) showing significantly higher demand.

Overall, the project demonstrates that data-driven approaches using tools like Python, Pandas, and Plotly can effectively analyze healthcare data and support better decision-making. The insights obtained can help healthcare administrators improve resource allocation, manage hospital capacity, and prepare for future demand. The findings contribute to the growing body of evidence supporting the integration of data analytics into healthcare management practices.

6.2 Future Enhancements

The following enhancements are proposed to extend the capabilities of the current system:

1. Real-Time Data Integration: The system can be enhanced by integrating real-time hospital data feeds for continuous monitoring and instant alerts when capacity thresholds are approached.
2. Machine Learning Predictions: Advanced machine learning techniques such as LSTM networks and Random Forest models can be implemented to predict future hospital and ICU demand based on historical patterns.
3. Web-Based Dashboard: A web-based or mobile-responsive dashboard can be developed using frameworks such as Dash or Flask for easier accessibility and remote usage.
4. Additional Data Variables: Patient demographics, disease patterns, seasonal trends, and socioeconomic factors can be incorporated for deeper and more contextual analysis.
5. Multi-Country Expansion: The system can be expanded to include data from multiple countries for global healthcare capacity analysis and benchmarking.
6. Automated Alerting System: Automation can be added to send real-time alerts to hospital administrators when hospital or ICU capacity exceeds critical threshold limits.
7. Natural Language Querying: Integration of natural language processing (NLP) capabilities would allow healthcare staff to query the dashboard using plain English questions.
8. Integration with Hospital Management Systems: Direct integration with existing Hospital Information Systems (HIS) would enable seamless data flow and reduce manual data preparation overhead.

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